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Wide-bandwidth Pound–Drever–Hall locking through a single-sideband modulator

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An integrated single-sideband modulator is used as the sole wide-bandwidth frequency actuator in a Pound–Drever–Hall locking loop. Thanks to the large modulation bandwidth, the device enables a locking range of ±75 MHz and a control bandwidth of 5 MHz without the need for a second feedback loop. As applied to the coupling of an extended-cavity diode laser at 1.55 μm to a high-finesse optical cavity, the in-loop frequency noise spectral density reaches a minimum of 1 mHz/Hz$^{1/2}$ at 1 kHz. © 2015 Optical Society of America

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Phase locking of laser fields to high-finesse optical cavities is a common approach to increase their short-term stability and to make them suitable for the interrogation of narrow or ultranarrow atomic or molecular transitions. When hertz or sub-hertz resolution is required [1–3], highly isolated and thermally stable cavities made up of ultralow-expansion (ULE) glass [4], or more recently of a silicon single crystal [5], are adopted. In several cases they are complemented by a simpler cavity with lower finesse for pre-stabilization and narrowing of the laser [6]. Optical cavities are also a common tool for less demanding applications such as Doppler-free molecular spectroscopy [7,8]: they allow the laser linewidth to be narrowed down to the few tens of kilohertz needed to resolve Lamb dips, at the same time enhancing the optical power density above the saturation threshold. In all cases, a simple and robust system providing tight locking of the laser to the cavity is highly desirable.

Pound–Drever–Hall (PDH) is one of the most powerful and common techniques [9,10] of laser frequency stabilization based on the interrogation of a cavity resonance by a phase modulated laser. The demodulated PDH error signal typically serves two proportional–integral/proportional–integral-derivative (PI/PID) control loops: one with high dynamic range to compensate for the large frequency fluctuations that occur at low Fourier frequencies, and another with high bandwidth to force the noisy laser field to follow that of the cavity mode. Piezoelectric transducers (PZTs) are typically employed for the first task, whereas fast control is typically achieved acting either on the laser current [11]—this is the typical approach with laser diodes that require control bandwidth in excess of 1 MHz—or on an external acousto-optic modulator [12], which is an ideal choice with low-noise solid-state lasers or fiber-Bragg-grating lasers.

In this Letter, we present a new implementation of the PDH technique based on a single-frequency actuator composed of an integrated lithium–niobate single-sideband modulator (SSM). A control bandwidth of 5 MHz, mainly limited by the modulation bandwidth of the voltage-controlled oscillator (VCO) at the output of the PID servo, complies here with a large locking range of about 150 MHz, thus making the lock stable for hours without any PZT. The approach is applied to the locking of an extended-cavity diode laser (ECDL) to a stainless steel high-finesse optical cavity with 3.7-kHz-large resonances and only a coarse suppression of acoustic, mechanical, and thermal perturbations: in such conditions, the in-loop laser-cavity relative frequency noise remains suppressed down to 1 mHz/Hz$^{1/2}$ below 1 kHz.

The experimental setup, as shown in Fig. 1, consists of a fibered ECDL, tunable in the 1.5–1.63 μm range, which injects a series of two integrated lithium–niobate electro-optic modulators. The first is a conventional phase modulator placing sidebands 16 MHz away from the carrier. The second is the optical equivalent of a single-sideband radiofrequency (RF) mixer, which is the device used in optical communications to imprint a four-level modulation in a quadrature-phase-shift-keying format. It is composed of two Mach–Zehnder modulators (MZMs) placed in the branches of a Mach–Zehnder interferometer (MZI). The carrier suppression needs the MZMs to be biased at a transmission minimum, whereas the suppression of one sideband needs a quadrature bias point for the MZI together with quadrature RF signals applied to the MZMs. When these conditions are met, the device synthesizes a single sideband that may be tuned by more than 20 GHz by simply changing the RF carrier: this may be particularly useful

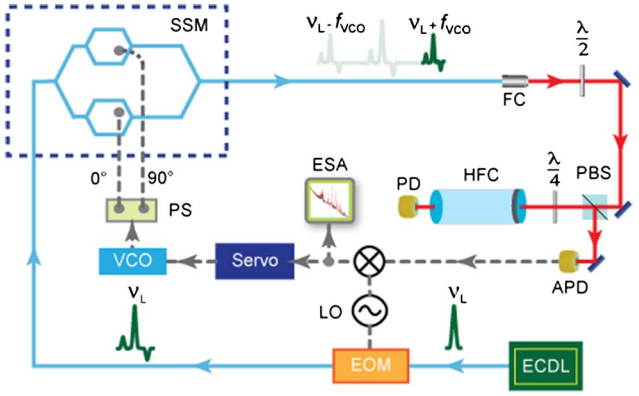

Fig. 1. Experimental setup. Blue line, optical fiber link; red line, free-space laser beam; dashed line, electrical signal; EOM, electro-optic modulator; HFC, high-finesse cavity; SSM, single-sideband modulator; PS, hybrid 90° phase-shifter; VCO, voltage-controlled oscillator; ESA, electrical spectrum analyzer; FC, fiber coupler; PBS, polarizing beam splitter; LO, local oscillator; PD, photodetector; APD, avalanche photodiode.

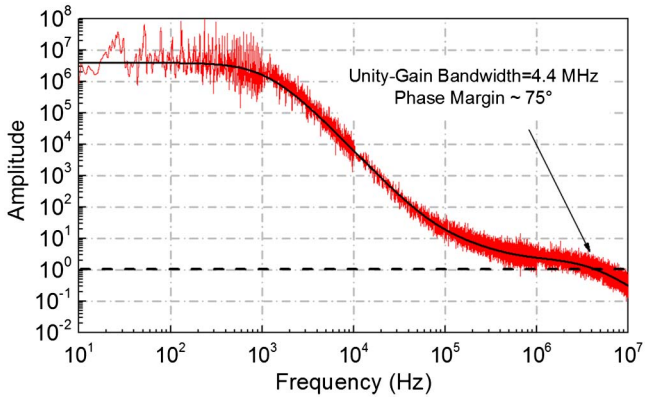

Fig. 2. Amplitude response of the PDH open-loop transfer function.

either to give tunability to an ultrastable laser locked to a ULE cavity [13] or to imprint ultrafast frequency corrections for locking the phase of a laser either to another laser [14] or to a high-finesse optical cavity, as hereafter demonstrated.

The fibered output of the SSM seeds through a variable focal-length collimator, a commercial stainless steel optical cavity with a free-spectral-range of 300 MHz (length of 50 cm) and a finesse of 80,000, implying 3.7-kHz resonances full-width at half-maximum (FWHM). To minimize the time lag in the control loop, a combination of beam splitter and quarter-waveplate has been preferred to an optical circulator for the extraction of the retroreflected signal for PDH locking. This is measured by an avalanche photodetector with a 50 MHz bandwidth and sent after demodulation to a PID controller whose output drives a VCO operating at a central frequency of 800 MHz. With a 22 MHz/V frequency-to-voltage conversion coefficient, the VCO allows tracking of cavity drifts as large as ±75 MHz. To keep the electronic part as simple as possible, the VCO output directly drives the SSM through a hybrid 90° phase-shifter operating from 500 MHz to 7 GHz without intermediate amplification. With a RF power of 6 dBm (against an optimum value of 18 dBm) and an input optical power of 18 mW, the SSM output amounts to ~100 μW, thus comparable to the power level often adopted with high-finesse etalons. The cavity transmission is monitored with an amplified InGaAs detector with a gain of 10^5 V/W and a bandwidth of 5 MHz.

Figure 2 shows the open-loop transfer function of the PDH system based on the experimental characterization of each loop component, namely PID, VCO, SSM, and PDH frequency discriminator. (For details on it, see below.) The PID, in particular, provides a flat 70-dB gain below 1 kHz, i.e., where the open-loop laser-cavity frequency noise is higher. It acts as a double integrator to a corner frequency of 70 kHz. There the gain returns flat until a zero is placed at 500 kHz, as compensated at higher frequencies by an upper pole at 4 MHz. The differential stage was added to maximize the unity-gain frequency of the loop, counteracting the low-frequency pole at 1.85 kHz (f_C, half of the cavity mode width) set by the cavity. In these conditions, a control bandwidth of 4.4 MHz is expected, with a phase margin of ~75°.

Figure 3 shows the time trace of the demodulated error signal (red trace) and of the transmitted power (green trace) when the cavity is detuned from the laser with a sawtooth ramp. The curves in Fig. 3(a) show the typical triple peak of the PDH error signal in an open-loop regime, while Fig. 3(b) reports a zoomed-in trace around the central feature. The error signal and the transmitted power quickly deviate from zero when the cavity mode starts superimposing the laser mode, whereas they slowly decrease when departing from it. The exponential decay is due to nonadiabatic injection conditions, while the optical fringes come from the interference between the fixed-frequency input radiation and the intra-cavity field [15,16], which is Doppler-shifted at every mirror bounce [17]. As shown in Fig. 3(c), when the proportional gain is activated, the feedback loop efficiently forces the laser to remain locked in spite of the quite fast frequency scan rate of ~800 MHz/s. The output power increases by a factor of 7 with respect to the

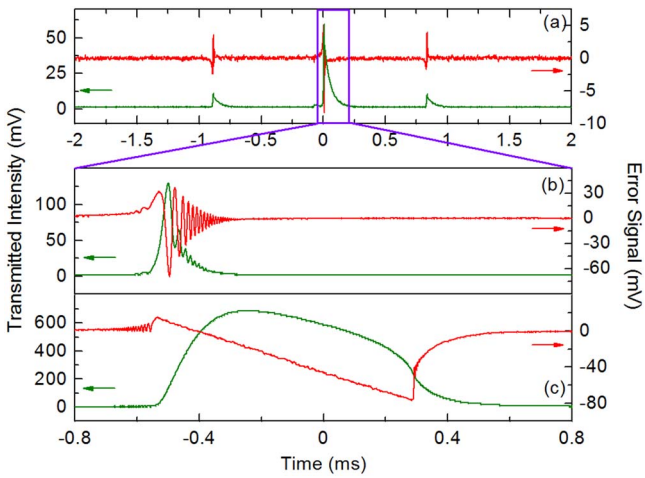

Fig. 3. Pound–Drever–Hall error signal (red curves, right scale) and power transmitted through the cavity (green curve, left scale) as the cavity-length is scanned. Panel (a) gives a wide-range view, whereas panels (b) and (c) refer to a zoomed-in window acquired at half the scan speed, without and with engaging the proportional servo, respectively.

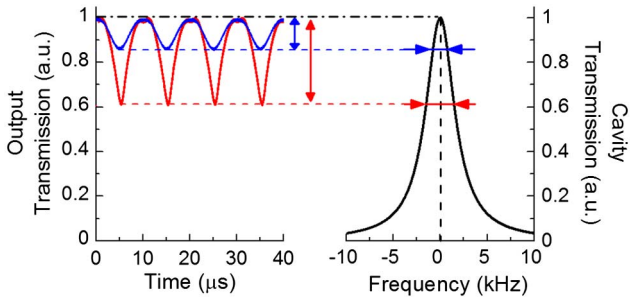

Fig. 4. Left: optical power at the cavity output upon dithering at 100 Hz the PDH working point by 90 mV (red curve) and 50 mV (blue curve). The power drop, as related to the cavity mode transmission (right curve), allows measuring the voltage-to-frequency slope of the PDH discriminator in a closed-loop regime.

open-loop regime and exhibits an almost flat behavior over a range that corresponds to about 600 kHz. Correspondingly, the PDH error signal is drawn down to the minimum of its dynamic, which corresponds to the point where the lock is lost.

For the calibration of the PDH frequency discriminator we preferred not to use the open-loop traces reported in Fig. 3 due to the nonadiabatic conditions. We preferred to modulate in a closed-loop regime the voltage set-point of the PDH servo while simultaneously acquiring the cavity transmission. The power drop at the cavity output can in fact be easily translated into a frequency shift once the cavity mode width is known. In our case, this was determined to be 3.7 kHz from the 43 μs cavity photon lifetime. Figure 4 reports the normalized cavity throughput under the application of two sinusoidal modulation signals at a frequency of 100 Hz. For modulation amplitudes falling within the linear response region of the discriminator, all tests converged toward a voltage-to-frequency conversion factor of $3.1 \cdot 10^5$ V/Hz.

Figure 5 reports the amplitude spectral density of the in-loop frequency noise, as retrieved from the voltage noise of the error signal using the above V/Hz conversion factor and after

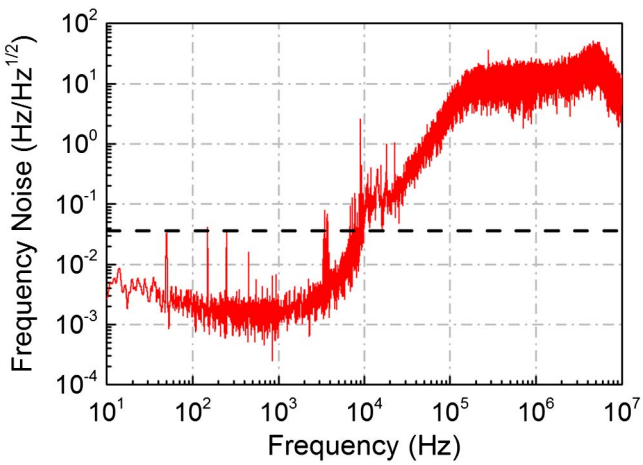

Fig. 5. Relative frequency noise spectral density between laser and cavity, as rescaled by $(1 + jf/f_C)$ to compensate the low-pass filtering effect given by the cavity. Black dashed line, photodetector shot-noise limit.

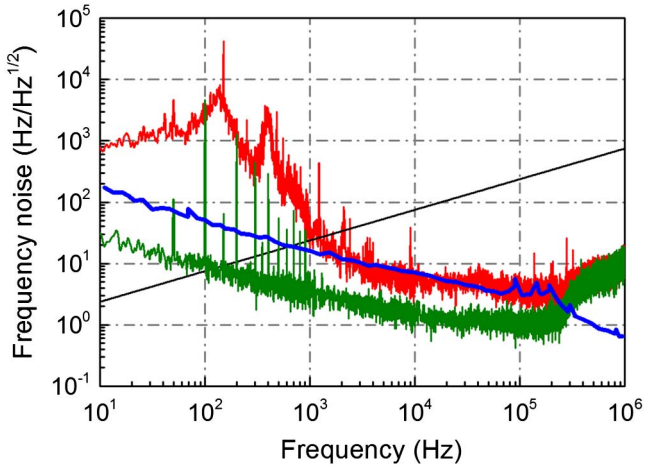

Fig. 6. Red curve, frequency noise spectral density of the cavity-stabilized laser against a slowly locked reference laser, as measured by means of a frequency-to-amplitude converter followed by an electrical spectrum analyzer; blue curve, frequency noise spectrum of the reference laser (from datasheet); green curve, noise floor of the frequency-to-amplitude converter; black curve, beta line [20].

rescaling by the cavity frequency response. (See caption for details.) Apart from a residual signal at 50 Hz and odd harmonics given by the power line, the frequency noise spectrum remains essentially flat up to about 1 kHz, which is the point where the gain loop starts to decrease steeply. The noise floor attains a remarkably low minimum of 1 mHz/Hz$^{1/2}$, which attests to the high noise suppression capability of the SSM-based PDH lock. On the low-frequency side of the noise spectral density, the slight increase below 100 Hz is due to the large 1/f noise of the cavity, which exceeds in such region the 10^3 Hz/Hz$^{1/2}$ level. (See Fig. 6 and relative comments below.) The noise floor remains, however, well below the limit of 36 mHz/Hz$^{1/2}$ (dashed line in Fig. 5) given by the photodetection shot-noise limit. As described, e.g., in [18], such noise (and in general the discriminator noise) cannot be observed at the error point, yet it ultimately determines the relative linewidth, down to a 4 mHz FWHM in our case. On the high-frequency side of the spectrum, the peak at about 10 kHz comes from the laser, whereas the servo bump at 5 MHz is the signature of a very high-control bandwidth, in good agreement with the estimation from data in Fig. 2 and favorably comparing to the best results reported in the literature for semiconductor lasers [11,19].

The limit to the bandwidth comes mainly from the 20-ns-level time delay introduced by the VCO, whose choice was mostly influenced in our case by dynamics constraints, namely, by the need to counteract the drifts of the unstabilized cavity. With more stable cavities, one could easily design a combination of VCO and PID parameters offering more bandwidth and/or a lower laser-cavity relative frequency noise. For the purpose of quantifying the amount of cavity noise, the cavity-stabilized laser was made to beat against a stable reference laser (NKT Koheras Adjustik E15 Custom) and the beat note signal was analyzed in the frequency domain with the help of a frequency-to-amplitude converter. The corresponding spectrum shown in Fig. 6 refers to conditions where a slow lock between the lasers with a nearly 100-Hz bandwidth was

introduced to counteract their drifts. As expected, the noise spectrum of the beat note closely follows the frequency noise of the master laser (from datasheet) above 2 kHz, i.e., where the technical noise of the cavity drops off. Below that threshold, the frequency noise of the cavity strongly dominates, with an intense peak at 400 Hz given by the mechanical resonance of the cavity. Its integration from the beta-line crossing point [20] down to 10 Hz gives an effective cavity linewidth as large as 60 kHz (over about 100 ms). Actually, this is an indicative yet rough evaluation, as the integral is overestimated at around 100 Hz due to the servo bump and underestimated below 100 Hz by the negative feedback given the slow lock. The main point is, however, that the unfavorable operating conditions do not prevent the laser from remaining steadily and robustly locked until the point where cavity thermal drifts exceed the 75 MHz dynamic range given by the PID–VCO combination.

In conclusion, we have shown that an integrated SSM may provide tight lock of a laser to a high-finesse cavity with a versatile and robust fiber-format arrangement that makes use of just one servo and one external actuator. In these conditions the laser is not perturbed, nor is a fast frequency modulation input port required. Due to the gigahertz bandwidth of the SSM, the 5-MHz bandwidth of our implementation does not represent a fundamental limit, but the result of a choice trading off bandwidth and dynamic range due to the adoption of an unstabilized cavity. Commercially available SSMs may operate from 1.4 to 1.65 μm, but extension to other ranges, from the visible to the mid-infrared, is technologically feasible. Relatively simple configurations based on lumped electrodes could be used, since laser stabilization does not require gigahertz modulation bandwidths. Moreover, the addition on the same chip of a phase modulator would provide a monolithic plug-and-play device for PDH locking, to be applied for the realization of sub-hertz linewidth lasers in combination with ultrastable cavities, or for simpler line-narrowing purposes with, e.g., monolithic etalons.

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